



Artificial Intelligence: An introduction

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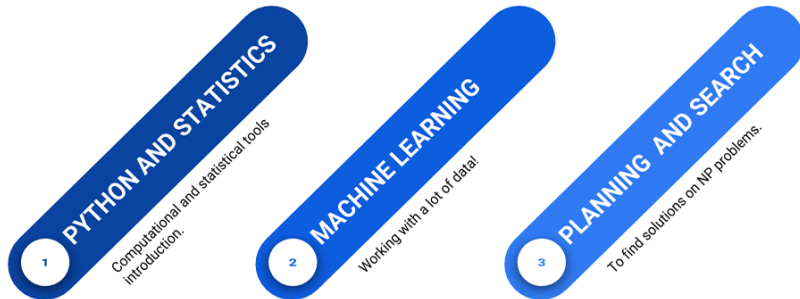


Biomedical Imaging, Vision and Learning Laboratory

Información del profesor

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- ▶ Contenido del curso:
<https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student>

Course outline



- ▶ **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- ▶ **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course



Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

First module: Python and statistics introduction

- ▶ Introduction to Python
- ▶ Introduction to statistics
- ▶ Introduction to Visualization and data analytics



Second Module: Machine Learning

Classification and regression

- ▶ Naive Bayes
- ▶ Classification trees
- ▶ Support Vector Machines
- ▶ Introduction to Deep-learning

Non-supervised learning

- ▶ K-means
- ▶ DBScan



Third module: Planning and Searching

- ▶ Planning and searching
- ▶ Genetic algorithms (GA)
- ▶ Simulated annealing (SA)



Some I.A. Applications: In classrooms



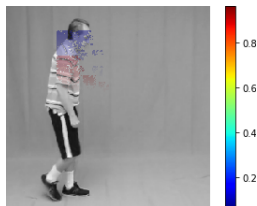
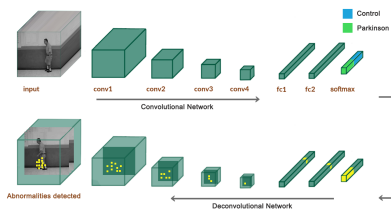
Video AI-classroom-CHINA



Some I.A. Applications: In Agriculture



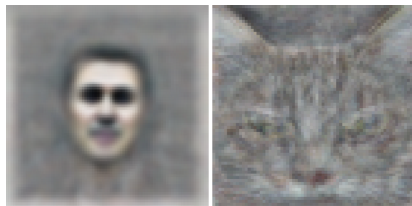
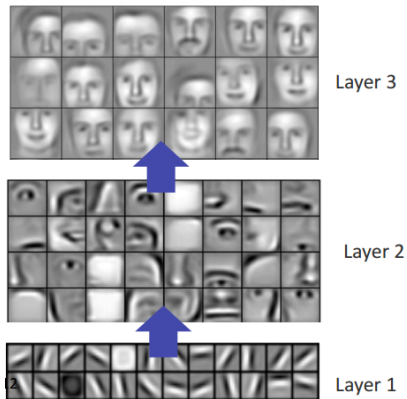
Salient Gait maps



Learning convolutional neuronal patterns

A 3D convolutional gait representation that produces salient spatio-temporal regions can potentially support and complement the diagnosis and the following of Parkinson's disease

Big challenges: How understand complex concepts?



Now, the machines can dream





A photograph of a Go match between AlphaGo and Lee Sedol. Two men are seated at a table with a Go board. The man on the left is AlphaGo, and the man on the right is Lee Sedol. The Go board is visible on the table. The background is dark.

Video 01

ALPHAGO

A photograph of a Go match between a human player and the AI program AlphaGo. The human player, wearing a dark suit, is seated on the left, looking at a board. AlphaGo, represented by a white board with black stones, is on the right. The board is on a stand with a blue base. The background is dark. The text "Video 02" is overlaid in green at the top right. The word "ALPHAGO" is written in large white letters at the bottom.

Video 02

ALPHAGO

ImageNet Challenge

IMAGENET

IMAGENET Netflix 22:30, Bill Nye

- 1,000 object classes (categories).
- Images:
 - 1.2 M train
 - 100k test.

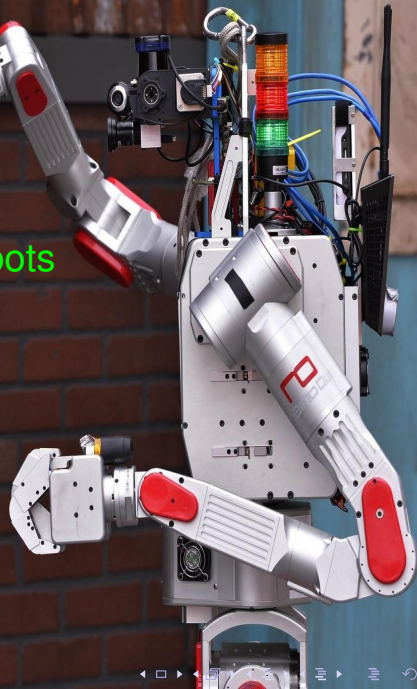


As support for decisions making

- ▶ Medicine
- ▶ Politics
- ▶ Laws



NOVA Rise of the Robots





y ahora ...Atlas. Google again!

Boston Dynamics

DARPA



Structure of the Course

- ▶ 40% Talleres (Problemsets)
- ▶ 30% Parciales (Quizes)
- ▶ 30% Proyecto funcional IA
 - ▶ Solo se evalua el curso, si ha sido previamente aprobado antes del primer corte.

Curso en GITLAB:

`https://gitlab.com/bivl2ab/academico/
cursos-uis/ai/ai-uis-student`

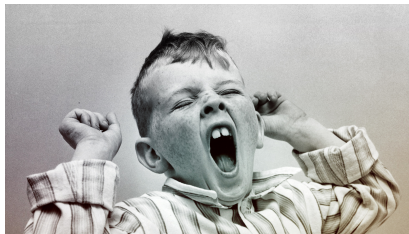
PREVIOUS PROJECTS

- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2018-2.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-1.md
- ▶ https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-2.md
https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2020-1.md

Understanding Artificial Intelligence and Its Future



Thank you for your attention ...



... It's time to wake up

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